

→ Suppose that the data mining task is to cluster the following eight points into 3 clusters:
The data for these objects is as

$A_1(2, 10)$

$A_2(2, 5)$

$A_3(8, 4)$

$B_1(5, 8)$

$B_2(7, 5)$

$B_3(6, 4)$

$C_1(1, 2)$

$C_2(4, 9)$

	x_1	x_2
A_1	2	10
A_2	2	5
A_3	8	4
B_1	5	8
B_2	7	5
B_3	6	4
C_1	1	2
C_2	4	9

Here, the data for is converted into 8 objects for clustering processes

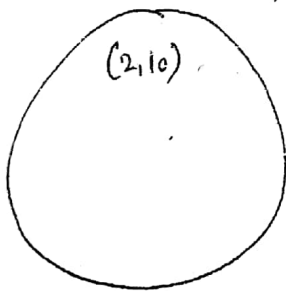
The distance function is Euclidean distance. Suppose initially we assign A_1 , B_1 , and C_1 as the center of each cluster, respectively. Use the k-means algorithm to show The Final 3 clusters

Given Centroids are A_1, B_1, C_1 Step-1: Initially A_1, B_1, C_1 are 3 clusters

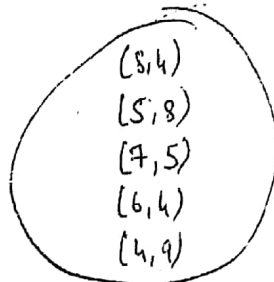
Step-2:

Point Centroids	(2, 10)	(2, 5)	(8, 4)	(5, 8)	(7, 5)	(6, 4)	(1, 2)	(4, 9)
C_1 (2, 10)	0	5	8.485	3.605	7.071	7.21	8.062	2.236
C_2 (5, 8)	3.605	4.242	5	0	3.605	4.123	7.21	1.414
C_3 (1, 2)	8.062	3.162	7.280	7.21	6.708	5.385	0	7.665

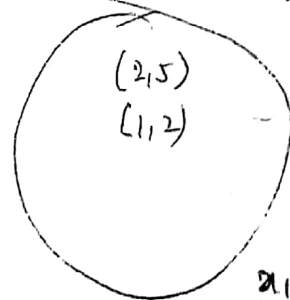
cluster-1 ~~(2, 5)~~



cluster-2 ~~(5, 8)~~



cluster-3 ~~(1, 2)~~



Step-3: Calculating Centroids.

Centroid	cluster
(2, 10)	cluster-1
(6, 6)	cluster-2
(4.5, 3.5)	cluster-3

data set

$A_1(5, 8, 9)$
 $A_2(7, 10, 12)$
 x_1, y_1, z_1
 x_2, y_2, z_2

A_1
 A_2
 A_3
 B_1
 B_2
 B_3
 C_1

		3 dim
		$A_1(5, 8, 9, 10)$ $A_2(7, 9, 8, 12)$
	2	10
	2	5
	8	4
	5	8

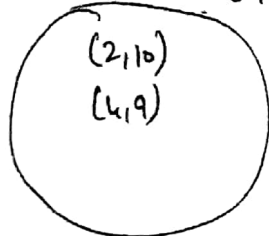
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Centroid \ Points	(2,10)	(2,5)	(8,4)	(5,8)	(7,5)	(6,4)	(1,2)	(4,9)
C_1 (2,10)	0	5	8.48	3.605	7.07	7.211	8.06	2.23
C_2 (6,6)	5.65	4.12	2.828	2.23	1.414	2	6.40	3.60
C_3 (1.5, 3.5)	6.51	1.58	6.51	5.70	5.70	4.52	1.58	6.041

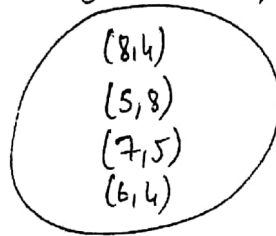
Step IV

Cluster	Centroid
1	(3, 9.5)
2	(6.5, 5.25)
3	(1.5, 3.5)

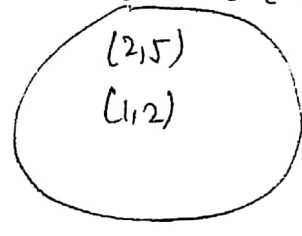
cluster-1 (~~(2,10)~~)



cluster-2 (~~(6,6)~~)

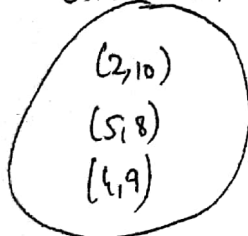


cluster-3 (~~(1,2)~~)

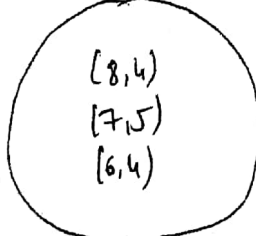


Centroid \ Points	(2,10)	(2,5)	(8,4)	(5,8)	(7,5)	(6,4)	(1,2)	(4,9)
C_1 (3, 9.5)	1.118	4.609	7.433	2.5	6.020	6.264	7.762	1.118
C_2 (6.5, 5.25)	6.54	4.506	1.95	3.13	0.559	1.346	6.388	4.506
C_3 (1.5, 3.5)	6.519	1.581	6.51	5.700	5.700	4.527	1.581	6.04

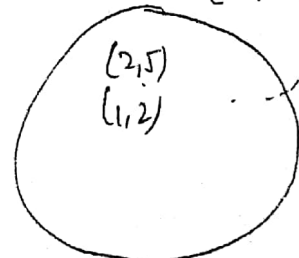
cluster-1 (~~(2,10)~~)



cluster-2 (~~(6,6)~~)



cluster-3 (~~(1,2)~~)

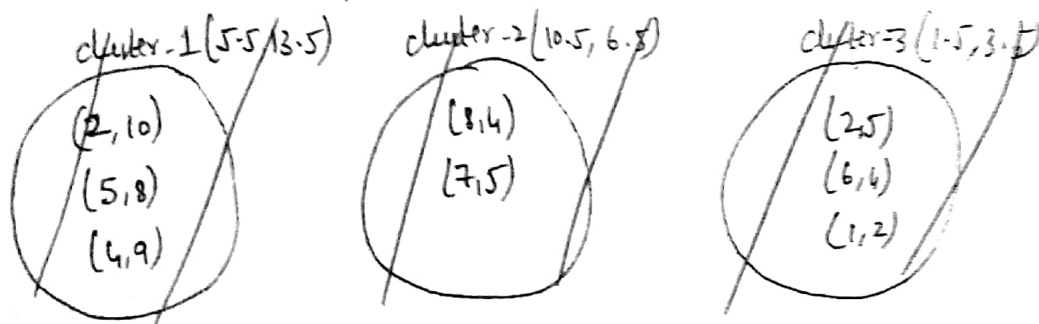


Step 5

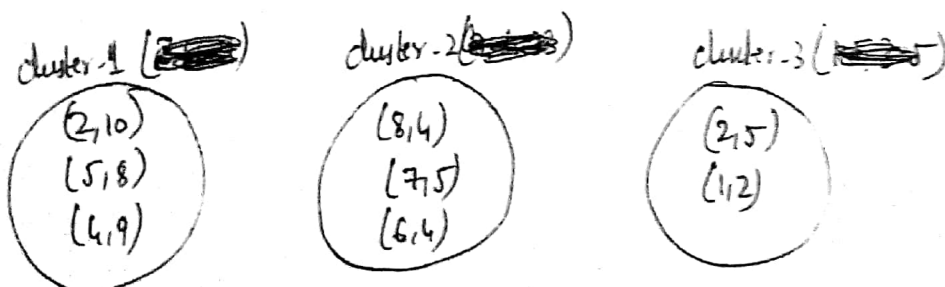
Cluster	Centroid
1	(3, 9.5) (3.66, 9)
2	(6.5, 5.25) (7, 4.33)
3	(1.5, 3.5) (1.5, 3.5)

Points \ Centroid	(2,10)	(2,5)	(8,4)	(5,8)	(7,5)	(6,4)	(1,2)	(4,9)
(5.5, 13.5)	4.94	9.19	9.82	5.52	8.63	9.51	12.3	6.7
(10.5, 6.5)	9.19	8.63	3.53	5.7	3.8	5.14	10.51	6.96
(1.5, 3.5)	6.519	1.58	6.51	5.700	5.700	4.527	1.58	6.04

Centroid	cluster
(3.66, 9)	1
(7, 4.33)	2
(1.5, 3.5)	3



Points \ Centroid	(2,10)	(2,5)	(8,4)	(5,8)	(7,5)	(6,4)	(1,2)	(4,9)
C1 (3.66, 9)	1.937	4.33	6.62	1.67	5.2	5.52	7.48	0.30
C2 (7, 4.33)	7.55	5.04	1.05	4.17	0.67	4.05	6.4	5.55
C3 (1.5, 3.5)	6.51	3.8	6.51	5.700	5.700	4.52	1.58	6.04



Step 6: Find the Centroids

cluster	Centroid
1	(3.66, 9)
2	(7, 4.33)

The Centroids of the clusters for step-5 & step-6 are same, to stop the procedure.

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Important points to Remember:-

- 1) In classification algorithms
i) There should be a target variable.
ii) Variables play an important role
- 2) In clustering algorithms
i) No Target Variable.
ii) Observations play an important role (objects)
iii) All the rows must be named

K-Means 2nd Example :-

Subject	A	B
1	1.0	1.0
2	1.5	2
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

The objects for the K-Means problem are as follows & Rows will be given names

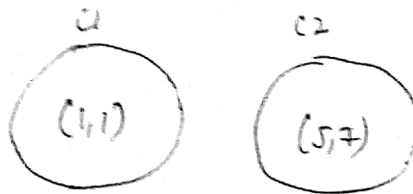
A1 (1.0, 1.0)
A2 (1.5, 2)
A3 (3.0, 4.0)
A4 (5.0, 7.0)
A5 (3.5, 5.0)
A6 (4.5, 5.0)
A7 (3.5, 4.5)

→ In K-Means Algorithm, we should select 'K' first. Here given that $K=2$

→ Given centroids are A1, A4

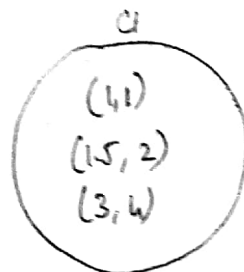
Step 1

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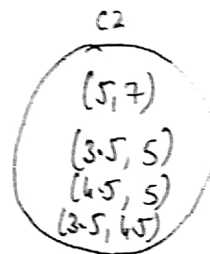


Step 2

Points \ Centroid	(1,1)	(1.5,2)	(3,4)	(5,7)	(3.5,5)	(4.5,5)	(3.5,4.5)
C1 (1,1)	0	1.1	3.60	7.2	4.7	5.3	4.3
C2 (5,7)	7.2	6.1	3.60	0	2.5	2.06	2.91



Centroid = (1.83, 2.33)

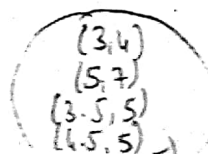
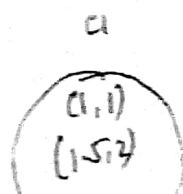


Centroid = (4.12, 5.37)

Centroids	Centroids
C1	(1.8, 2.3)
C2	(4.1, 5.4)

Step 3

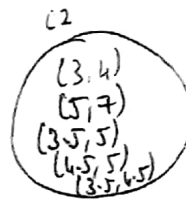
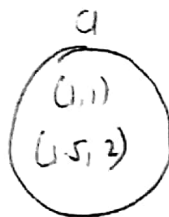
Points \ Centroid	(1,1)	(1.5,2)	(3,4)	(5,7)	(3.5,5)	(4.5,5)	(3.5,4.5)
(1.8, 2.3)	1.5	0.4	2.0	5.6	3.1	3.7	2.7
(4.1, 5.4)	5.3	4.2	1.7	1.8	0.7	0.5	1.0



clusters	Centroid
c1	(1.25, 1.5)
c2	(3.9, 5.1)

Step-4

Points Centroid	(1,1)	(1.5, 2)	(3,4)	(5,7)	(3.5,5)	(4.5,5)	(3.5,4.5)
(1.25, 1.5)	0.5	0.5	3	6.6	4.1	4.7	3.7
(3.9, 5.1)	5.0	3.9	1.4	2.1	0.4	0.6	0.7



clusters	centroid
c1	(1.25, 1.5)
c2	(3.9, 5.1)

The Centroids of the clusters for step-3 & step-4 are same. So stop the procedure

Now to Assign the objects into clusters for each iteration (from the table)

- 1) find the Euclidean distance between the objects & the cluster centroids.
- 2) The object is assigned to the cluster in which the distance b/w the object & the cluster centroid is minimum.
- 3) Similarly, apply the same procedure for all the objects

When we stop the clustering process -

- 1) the Centroids of the clusters of two consecutive iterations are same, then we can stop the procedure.

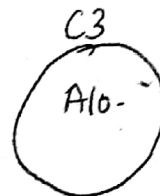
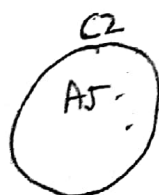
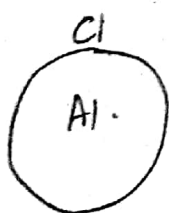
K-Modes Problem :-

(71)

RID	Age	Income	Student	Credit-rating
A1	Youth	High	No	Fair
A2	Youth	High	No	Excellent
A3	Middle-aged	High	No	Fair
A4	Senior	Medium	No	Fair
A5	Senior	Low	Yes	Fair
A6	Senior	Low	Yes	Excellent
A7	Middle-aged	Low	Yes	Excellent
A8	Youth	Medium	No	Fair
A9	Youth	Low	Yes	Fair
A10	Senior	Medium	Yes	Fair
A11	Youth	Medium	Yes	Excellent
A12	Middle-aged	Medium	No	Excellent
A13	Middle-aged	High	Yes	Fair
A14	Senior	Medium	No	Excellent

Centroid - $\frac{x_1 + x_2 + x_3}{3}$
mode - highest frequency

Given $k=3$



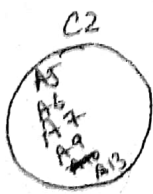
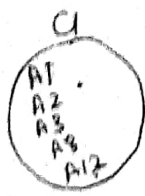
Same - Similar - 0
youth - youth - 0
not same - 1

Here we will find the centers of clusters using Mode Concept

Centers	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12	A13	A14
A1	0	1	1	2	3	4	4	1	2	3	3	3	2	3
A5	3	4	3	2	0	1	2	3	1	1	3	4	2	3
A10	3	4	3	1	1	2	3	2	2	0	2	3	2	2

(Youth, High, no, Fair)

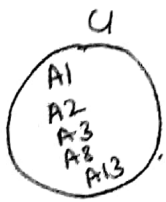
1. Find the distance using MINIMUM SIMILARITY MEASURE



→ finding the Center using Mode

C1 = Youth High No Fair
 C2 = ~~Low~~ ~~Low~~ Yes Fair
 C3 = Senior Medium ~~No~~ ~~Fair~~ Excellent

observed Center	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12	A13	A14
C1 (Y, H, N, F)	0	1	1	2	3	4	4	1	2	3	3	3	2	3
C2 (S, L, Y, F)	3	4	3	2	0	1	2	3	1	1	3	4	2	3
C3 (S, M, N, E)	3	2	3	1	3	2	3	2	4	2	2	1	4	0



C1 = Youth High No Fair
 C2 = Senior low Yes Fair
 C3 = Senior Medium No Excellent

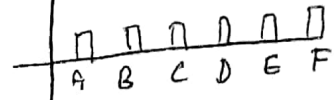
Stopping Criterion: 1. K modes Centers are equal
 2. cluster Points same
K means 1. cluster Points & cluster centroid are same

HIERARCHICAL CLUSTERING :- (SINGLE LINKAGE $\rightarrow$ MIN FUNCTION)

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1)

	A	B	C	D	E	F
A	0.00	0.71	5.66	3.61	4.24	3.20
B	0.71	0.00	4.95	2.92	3.54	2.50
C	5.66	4.95	0.00	2.24	1.41	2.50
D	3.61	2.92	2.24	0.00	1.00	0.50
E	4.24	3.54	1.41	1.00	0.00	1.12
F	3.20	2.50	2.50	0.50	1.12	0.00



$\rightarrow$ In the above distance table, the minimum value is 0.50, so the clusters D and F are very near, so we can club these two clusters and make one cluster (D,F).

$\rightarrow$ Now the clusters decreases to 5

	A	B	C	(D,F)	E
A	0.00	0.71	5.66	3.20	4.24
B	0.71	0.00	4.95	2.50	3.54
C	5.66	4.95	0.00	2.24	1.41
G _(D,F)	3.20	2.50	2.24	0.00	1.00
E	4.24	3.54	1.41	1.00	0.00

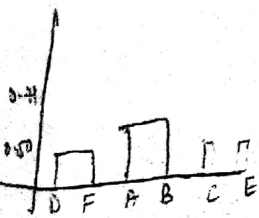


$$d(D,F) \rightarrow A = \min(d(D \rightarrow A), d(F \rightarrow A)) = \min(3.61, 3.20) = 3.20$$

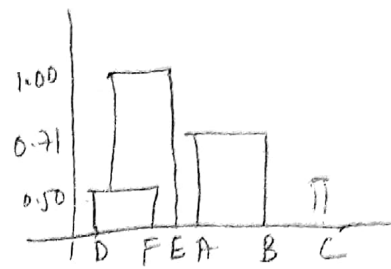
	(A,B)	C	(D,F)	E
(A,B)	0	4.95	2.50	3.54
C	4.95	0	2.24	1.41
(D,F)	2.50	2.24	0	1.00
G _(A,B)	3.54	1.41	1.00	0

$$\rightarrow d((A,B), (D,F)) =$$

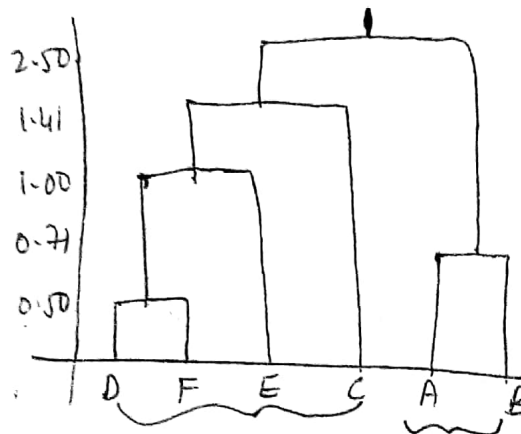
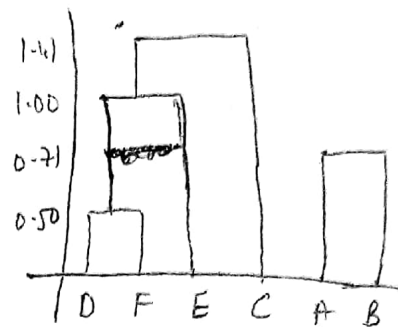
$$\min(A \rightarrow D, B \rightarrow D, A \rightarrow F, B \rightarrow F) \\ \min(3.61, 2.92, 3.20, 2.50) \\ 2.50$$



	(A,B)	C	(D,E,F)
(A,B)	0	4.95	2.50
C	4.95	0	(1.41)
(D,E,F)	2.50	(1.41)	0



	(A,B)	(C,D,E,F)
(A,B)	0	(2.50)
(C,D,E,F)	(2.50)	0



2) COMPLETE LINKAGE HIERARCHICAL CLUSTER $\rightarrow$ MAX () FUNCTION 75

Dissimilarity matrix

	A	B	C	D	E	F
A	0.00	0.71	5.66	3.61	4.24	3.20
B	0.71	0.00	4.95	2.92	3.54	2.50
C	5.66	4.95	0.00	2.24	1.41	2.50
D	3.61	2.92	2.24	0.00	1.00	0.50
E	4.24	3.54	1.41	1.00	0.00	1.12
F	3.20	2.50	2.50	0.50	1.12	0.00

① $\rightarrow$ We are finding the Dissimilarity Matrix

$\rightarrow$ Distance Matrix is nothing but Dissimilarity Matrix

$\rightarrow$ In the Dissimilarity Matrix, If the two objects are having less dissimilarity (min) means the two objects are more similar

② After Merging the clusters in each step, to find the table values for the new merged clusters, we have to use 3 concepts

	A	B	C	(D,F)	E
A	0.00	0.71	5.66	3.61	4.24
B	0.71	0.00	4.95	2.92	3.54
C	5.66	4.95	0.00	2.50	1.41
(D,F)	3.61	2.92	2.50	0.00	1.12
E	4.24	3.54	1.41	1.12	0.00

D F

use max function

$d(D,F) \rightarrow A$

$\Rightarrow \max(d(D,A), d(F,A))$

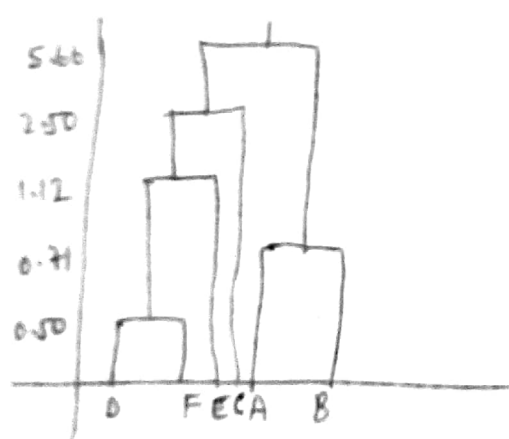
$= \max(3.61, 3.20)$

$= 3.61$

	(A,B)	C	(D,F)	E
(A,B)	0.00	5.66	3.61	4.24
C	5.66	0.00	2.50	1.41
(D,F)	3.61	2.50	0.00	1.12
E	4.24	1.41	1.12	0.00

	(A,B)	C	(D,E,F)
(A,B)	0	5.66	4.24
C	5.66	0	(2.50)
(A,E,F)	4.24	(2.50)	0

	(A,B)	(C,D,E,F)
(A,B)	0	(5.66)
(C,D,E,F)	(5.66)	0



~~Relay La Brouche~~

3). Average linkage hierarchical clustering $\rightarrow$ Avg() function

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	A	B	C	D	E	F
A	0.00	0.71	5.66	3.61	4.24	3.20
B	0.71	0.00	4.95	2.92	3.54	2.50
C	5.66	4.95	0.00	2.24	1.41	2.50
D	3.61	2.92	2.24	0.00	1.00	0.50
E	4.24	3.54	1.41	1.00	0.00	1.12
F	3.20	2.50	2.50	0.50	1.12	0.00

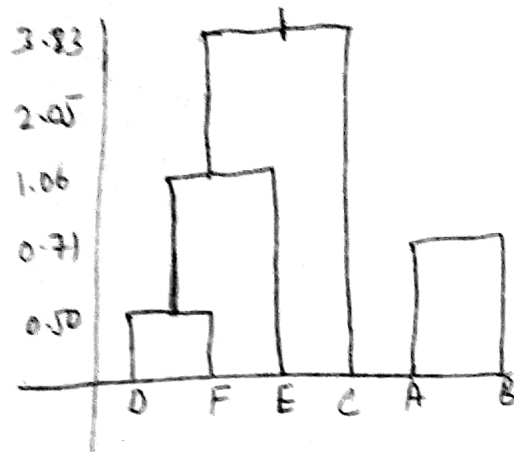
$$\begin{aligned}
 d(D, F) &\Rightarrow A \\
 &\Rightarrow \text{avg}(d(D, A), d(F, A)) \\
 &= \text{avg}(3.61, 3.20) \\
 &= \frac{3.61 + 3.20}{2} \\
 &= 3.40
 \end{aligned}$$

	A	B	C	(D, F)	E
A	0.00	0.71	5.66	3.40	4.24
B	0.71	0.00	4.95	2.71	3.54
C	5.66	4.95	0.00	2.37	1.41
(D, F)	3.40	2.71	2.37	0.00	1.06
E	4.24	3.54	1.41	1.06	0.00

	(A, B)	C	(D, F)	E
(A, B)	0	5.30	3.05	3.89
C	5.30	0	2.37	1.41
(D, F)	3.05	2.37	0	1.06
E	3.89	1.41	1.06	0

	(A, B)	C	(D, E, F)
(A, B)	0	5.30	3.33
C	5.30	0	2.05
(D, E, F)	3.33	2.05	0

	(A,B)	(C,D,G,F)
(A,B)	0	3.83
(C,D,G,F)	3.83	0



K-Prototypes:-

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	Department	Status	Age	Weight	Count
A1	Sales	Senior	31	46	30
A2	Sales	Junior	26	26	40
A3	Sales	Junior	31	31	40
A4	Systems	Junior	21	46	20
A5	Systems	Senior	31	66	5
A6	Systems	Junior	26	46	30
A7	Systems	Senior	41	66	3
A8	Marketing	Senior	36	46	10
A9	Marketing	Junior	31	41	4
A10	Secretary	Senior	46	36	4
A11	Secretary	Junior	26	26	6

Procedure 1 Find

S.D for each numerical attribute

2. average & find S.D's of the all numerical attributes

3 distance

$$d(A1, A2) =$$

For numerical attr

$$\frac{(x_1 - \bar{x}_1)^2 + (y_1 - \bar{y}_1)^2 + (z_1 - \bar{z}_1)^2}{3} + \text{avg(SD's of num)} \times \text{MATCHING}$$

$$7.23 + 13.67 + 14.95 = \frac{35.85}{3}$$

$$\text{avg(SD's)} = 11.95$$

Similarity measure of categories attributes

$$SD(\text{Age}) = 7.23$$

$$SD(\text{Weight}) = 13.67$$

$$SD(\text{Count}) = 14.95$$

Prototype	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11
A1	0	560.85	360.85	271.7	1060.85	825.7	1264.85	460.85	772.7	1036.85	1072.7
A5	1060.85	2921.7	2521.7	760.85	0	464.85	106	485.85	697.7	1161.85	1697.7
A10	1036.85	1867.7	1617.7	1052.7	1161.85	572.7	961.85	271.85	321.7	0	539.85

$$d(A1 \rightarrow A1) = 0$$

$$d(A1 \rightarrow A2) = (31-26)^2 + (46-26)^2 + (30-40)^2 + 11.95 = 560.85$$

$$d(A1 \rightarrow A3) = (31-31)^2 + (46-31)^2 + (30-40)^2 + 11.95 = 360.85$$

some mismatches between

match - 0
unmatch - 1

K Prototypes = K means + K model
Combination of numerical & categorical attributes
Euclidean distance for numerical attributes

matching similarity measure

$$d(A1 \rightarrow A4) = (31-21)^2 + (46-44)^2 + (20-20)^2 + \frac{220.85}{11.95} (2)$$

$$= 271.7$$

$$d(A1 \rightarrow A5) = (31-31)^2 + (46-66)^2 + (30-5)^2 + \frac{220.85}{11.95} (1)$$

$$= 1060.85$$

$$d(A1 \rightarrow A6) = (31-26)^2 + (46-46)^2 + (30-3)^2 + \frac{220.85}{11.95} (2)$$

$$= 825.7$$

$$d(A1 \rightarrow A7) = (31-41)^2 + (46-66)^2 + (30-3)^2 + \frac{220.85}{11.95} (1)$$

$$= 1264.85$$

$$d(A1 \rightarrow A8) = (31-36)^2 + (46-46)^2 + (30-16)^2 + \frac{220.85}{11.95} (1)$$

$$= 1036.85$$

$$d(A1 \rightarrow A9) = (31-31)^2 + (46-41)^2 + (30-4)^2 + \frac{220.85}{11.95} (2)$$

$$= 772.7$$

$$d(A1 \rightarrow A10) = (31-46)^2 + (46-36)^2 + (30-4)^2 + \frac{220.85}{11.95} (1)$$

$$= 1036.85$$

$$d(A1 \rightarrow A11) = (31-26)^2 + (46-26)^2 + (30-6)^2 + \frac{220.85}{11.95} (2)$$

$$= 1072.7$$

$$d(A5 \rightarrow A1) = (31-31)^2 + (66-46)^2 + (5-30)^2 + \frac{220.85}{11.95} (1)$$

$$= 1060.85$$

$$d(A5 \rightarrow A2) = (31-26)^2 + (66-26)^2 + (5-40)^2 + \frac{220.85}{11.95} (2)$$

$$= 2921.7$$

$$d(A5 \rightarrow A3) = (31-31)^2 + (66-31)^2 + (5-40)^2 + \frac{220.85}{11.95} (2)$$

$$= 2521.7$$

$$d(A5 \rightarrow A4) = (31-21)^2 + (66-46)^2 + (5-20)^2 + \frac{220.85}{11.95} (1)$$

$$= 760.85$$

$$d(A5 \rightarrow A5) = 0$$

$$d(A5 \rightarrow A6) = (31-26)^2 + (66-46)^2 + (5-3)^2 + \frac{220.85}{11.95} (1)$$

$$= 464.85$$

$$d(A5 \rightarrow A7) = (31-41)^2 + (66-66)^2 + (5-3)^2 + \frac{220.85}{11.95} (0)$$

$$= 104$$

$$d(A5 \rightarrow A8) = (31-34)^2 + (66-46)^2 + (5-10)^2 + \cancel{2885}(1) \\ = 685.85$$

$$d(A5 \rightarrow A9) = (31-31)^2 + (66-41)^2 + (5-4)^2 + \cancel{3628}(2) \\ = 697.7$$

$$d(A5 \rightarrow A10) = (31-46)^2 + (66-36)^2 + (5-4)^2 + \cancel{3645}(1) \\ = 1161.85$$

$$d(A5 \rightarrow A11) = (31-26)^2 + (66-26)^2 + (5-6)^2 + \cancel{3585}(2) \\ = 1697.7$$

$$d(A6 \rightarrow A1) = (46-31)^2 + (36-46)^2 + (4-30)^2 + \cancel{3885}(1) \\ = 1036.85$$

$$d(A6 \rightarrow A2) = (46-26)^2 + (36-26)^2 + (4-40)^2 + \cancel{3825}(2) \\ = 1867.7$$

$$d(A6 \rightarrow A3) = (46-31)^2 + (36-31)^2 + (4-40)^2 + \cancel{3825}(2) \\ = 1617.7$$

$$d(A6 \rightarrow A4) = (46-21)^2 + (36-46)^2 + (4-20)^2 + \cancel{3825}(2) \\ = 1052.7$$

$$d(A6 \rightarrow A5) = (46-31)^2 + (36-66)^2 + (4-5)^2 + \cancel{3825}(1) \\ = 1161.85$$

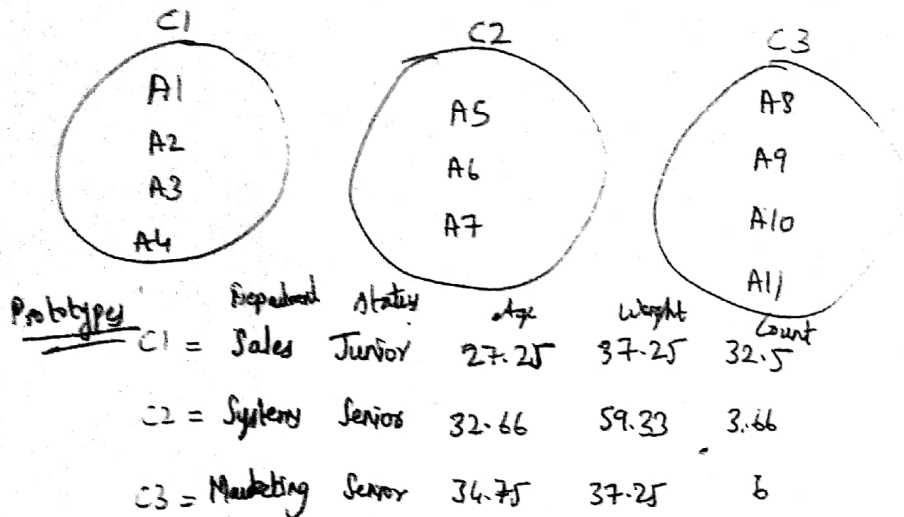
$$d(A6 \rightarrow A6) = (46-26)^2 + (36-46)^2 + (4-3)^2 + \cancel{3825}(2) \\ = 572.7$$

$$d(A6 \rightarrow A7) = (46-41)^2 + (36-66)^2 + (4-3)^2 + \cancel{3825}(1) \\ = 961.85$$

$$d(A6 \rightarrow A8) = (46-36)^2 + (36-46)^2 + (4-10)^2 + \cancel{3825}(1) \\ = 271.85$$

$$d(A6 \rightarrow A9) = (46-31)^2 + (36-41)^2 + (4-4)^2 + \cancel{3825}(2) \\ = 321.7$$

$$d(A6 \rightarrow A11) = (46-26)^2 + (36-26)^2 + (4-6)^2 + \cancel{3825}(1) \\ = 539.85$$



objects / Prototype	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11
C1	109.095	184.375	109.375	230.695	1613.315	956.595	1902.315	675.115	868.595	1181.915	938.595
C2	134.527	2483.867	2158.867	587.327	69.047	229.667	114.327	235.627	352.147	728.367	1174.367
C3	623.265	1372.605	1222.605	474.905	848.265	195.605	881.265	94.125	387.65	138.765	216.605

→ Here, Instead of taking the entire Data Set for finding σ (SD) i.e. the average of the SD of all the Numerical attributes.

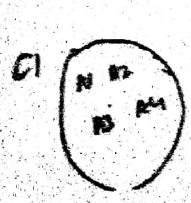
→ In this step we calculate σ for each cluster by taking the objects of the particular cluster

→ find σ for C

Here σ is nothing but σ

Here σ is average of the SD's of the Numerical Variables

for age, weight, Count for the objects



	age	weight	Count
A1	31	46	30
A2	26	26	40
A3	31	31	40

- i. $SD(\text{age}) = 4.79$
- ii. $SD(\text{weight}) = 10.30$
- iii. $SD(\text{Count}) = 9.57$



$AB = 7.64$
 $BC = 11.54$
 $CD = 1.08$
 $DA = 7.8$



$AB = 1.25$
 $BC = 9.54$
 $CD = 2.22$
 $DA = 6.64$

$$A(0+0) = (2+0)^2 + (3+2)^2 + (2+5-3)^2 + 1.12(1)$$

$$= 11.08$$

$$B(0+0) = (3+2)^2 + (3+2)^2 + (3+5-2)^2 + 1.12(1)$$

$$= 19.39$$

$$C(0+0) = (3+2)^2 + (3+2)^2 + (3+5-2)^2 + 1.12(1)$$

$$= 19.39$$

$$D(0+0) = (3+2)^2 + (3+2)^2 + (3+5-2)^2 + 1.12(1)$$

$$= 19.39$$

$$A(0+0) = (3+2)^2 + (3+2)^2 + (3+5-2)^2 + 1.12(2)$$

$$= 19.39$$

$$B(0+0) = (3+2)^2 + (3+2)^2 + (3+5-2)^2 + 1.12(1)$$

$$= 19.39$$

$$C(0+0) = (3+2)^2 + (3+2)^2 + (3+5-2)^2 + 1.12(2)$$

$$= 19.39$$

$$D(0+0) = (3+2)^2 + (3+2)^2 + (3+5-2)^2 + 1.12(2)$$

$$= 19.39$$

$$A(0+0) = (3+2)^2 + (3+2)^2 + (3+5-2)^2 + 1.12(1)$$

$$= 19.39$$

$$B(0+0) = (3+2)^2 + (3+2)^2 + (3+5-2)^2 + 1.12(2)$$

$$= 19.39$$

$$C(0+0) = (3+2)^2 + (3+2)^2 + (3+5-2)^2 + 1.12(1)$$

$$= 19.39$$